

The Arithmetic-Mean-at-Infinity Cancellation Envelope of Every Principal Operator Ideal Is Finite Rank

Candidate full solution packet for arXiv:0707.3271, Question 7

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Status: full solution, likely valid; pending specialist review

1 Source problem

Kaftal and Weiss define, for an operator ideal I ,

$$\widehat{I}^\infty := \bigcap \{J : J_{a_\infty} \supset I_{a_\infty}\},$$

and ask the following in Question 7 of their survey [1], on PDF page 24.

If $I = (\xi)$, is $\widehat{I}^\infty$ principal? If yes, is there an expression for its generator similar to the one in Definition 5.15 and with similar properties?

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Proposition 5.19. [36] Proposition 6.17]

If $F \neq (\xi) = (\widehat{\xi})$ with $\{0\} \neq \xi \in c_o^$, then $(\xi) = (\xi)_a$.*

There are principal ideals $I = \widehat{I}$ with $I \neq F$.

Example 5.20. [36] Example 6.18] *Let $\xi_j = \frac{1}{k}$ for $j \in ((k-1)!, k!]$. Then $(\xi) = (\widehat{\xi})$.*

Thus the condition $\xi \asymp \widehat{\xi}$ is strictly stronger than regularity.

While the definition of $\widehat{I}$ can be extended naturally to the am- ∞ case by setting $\widehat{I}^\infty := \bigcap \{J \mid J_{a_\infty} \supset I_{a_\infty}\}$, we have not analyzed its properties.

Question 7. If $I = (\xi)$, is $\widehat{I}^\infty$ principal? If yes, is there an expression for its generator similar to the one in Definition 5.15 and with similar properties?

Figure 1: Question 7 and its defining context in [1, p. 24].

2 Notation and source facts

Let c_0^* be the cone of nonnegative nonincreasing null sequences and let $(\ell^1)^*$ be its summable part. For $m \in \mathbb{N}$, the ampliation D_m repeats every entry m times, so

$$(D_m \eta)_n = \eta_{\lfloor n/m \rfloor}.$$

For $\eta \in (\ell^1)^*$, write

$$(\eta_{a_\infty})_n = \frac{1}{n} \sum_{j=n+1}^{\infty} \eta_j.$$

If (η) is the principal ideal generated by η , Calkin's characteristic-set criterion, in the notation of the source, is

$$\rho \in \Sigma((\eta)) \iff \rho \leq CD_m \eta \text{ for some } C > 0, m \in \mathbb{N}. \quad (1)$$

We use the following identity of Kaftal and Weiss [2, Lemma 4.7]. For an infinite-rank $\xi \in c_0^*$,

$$(\xi)_{a_\infty} = \begin{cases} (\xi_{a_\infty}), & \xi \in \ell^1, \\ \text{se}(\omega), & \xi \notin \ell^1, \end{cases} \quad (2)$$

where $\omega = \langle 1/n \rangle$. The finite-rank case will be handled directly. Finally, F denotes the ideal of finite-rank operators, the least nonzero ideal of $B(H)$.

3 Proof intuition

The key observation is that domination by an arithmetic mean at infinity is a tail-mass condition, not a pointwise lower bound. A fixed amount of tail mass may be spread over an arbitrarily long constant block, forcing that block's height to be as small as desired.

Given any infinite-rank test sequence ρ , we exploit this freedom diagonally. In the summable case we replace consecutive blocks of the source generator ξ by their averages. This preserves each block's mass and increases every intermediate tail, but the blocks can be chosen long enough that their heights defeat every fixed ampliation D_m in (1). In the nonsummable case, long low plateaux with mass at least one per block give a nonsummable generator, so (2) makes its arithmetic-mean-at-infinity ideal equal to the same universal ideal $\text{se}(\omega)$; the same diagonal argument excludes ρ . Thus every infinite-rank sequence is absent from at least one ideal in the defining intersection.

4 The classification

Theorem 1. *For every nonzero principal ideal I of $B(H)$,*

$$\boxed{\widehat{I}^\infty = F.}$$

Consequently $\widehat{I}^\infty$ is principal, independently of the generator of I , and any nonzero finite-rank operator is a generator.

Proof. First, $I_{a_\infty} \neq \{0\}$ for nonzero principal I . Hence every J in the defining intersection for $\widehat{I}^\infty$ is nonzero. Since F is contained in every nonzero operator ideal,

$$F \subset \widehat{I}^\infty. \quad (3)$$

We prove the reverse inclusion by showing that each infinite-rank $\rho \in c_0^*$ is omitted by some qualifying principal ideal J . Because ρ is nonincreasing and has infinite rank, $\rho_n > 0$ for every n .

Case 1: $I = F$. We first note that $F_{a_\infty} = F$. One inclusion is immediate because the arithmetic mean at infinity of a finite-support sequence again has finite support. Conversely, if α is supported on $\{1, \dots, r\}$, take $\zeta = \underbrace{\langle 1, \dots, 1, 0, \dots \rangle}_{r+1}$. Then $(\zeta_{a_\infty})_n = (r+1-n)/n > 0$ for $1 \leq n \leq r$, so

$\alpha \leq C\zeta_{a_\infty}$ for a suitable C . Thus $\alpha \in \Sigma(F_{a_\infty})$. It follows that $J = F$ itself belongs to the defining family, and therefore $\widehat{F}^\infty = F$.

Case 2: $I = (\xi)$, where $\xi \in (\ell^1)^*$ has infinite rank. Set $N_0 = 0$. Inductively, once N_{k-1} is chosen, put $q_k = N_{k-1} + 1$ and choose $N_k > N_{k-1}$ so large that

$$h_k := \frac{\sum_{j=q_k}^{N_k} \xi_j}{N_k - N_{k-1}} < \frac{\rho_{kq_k}}{k}. \quad (4)$$

This is possible because the numerator is bounded by $\|\xi\|_1$ while the denominator may be arbitrarily large. Define

$$\eta_j = h_k \quad (q_k \leq j \leq N_k).$$

The sequence η is nonincreasing: the average on a later block is no larger than its first ξ -entry, which is no larger than the last ξ -entry of the preceding block, which in turn is no larger than the preceding block average. Moreover, block masses are preserved, so $\eta \in (\ell^1)^*$ and $\sum_j \eta_j = \sum_j \xi_j$.

For $N_{k-1} \leq n \leq N_k$, the suffix $\xi_{n+1}, \dots, \xi_{N_k}$ has average no larger than the average of the whole k -th block, because ξ is nonincreasing. Therefore

$$\sum_{j=n+1}^{N_k} \eta_j = (N_k - n)h_k \geq \sum_{j=n+1}^{N_k} \xi_j.$$

All later block masses agree exactly. Hence, for every n ,

$$\sum_{j>n} \eta_j \geq \sum_{j>n} \xi_j, \quad \text{and thus} \quad \eta_{a_\infty} \geq \xi_{a_\infty}. \quad (5)$$

Let $J = (\eta)$. From (2) and (5),

$$J_{a_\infty} = (\eta_{a_\infty}) \supset (\xi_{a_\infty}) = I_{a_\infty},$$

so J qualifies for the defining intersection.

We claim that $\rho \notin \Sigma(J)$. Otherwise, by (1), $\rho \leq CD_m \eta$ for some $C > 0$ and $m \in \mathbb{N}$. Choose an integer $k \geq m$ with $k > C$ and evaluate at $n = mq_k$. Since $(D_m \eta)_{mq_k} = \eta_{q_k} = h_k$, monotonicity of ρ and (4) give

$$\rho_{mq_k} \geq \rho_{kq_k} > kh_k > C(D_m \eta)_{mq_k},$$

contradicting the proposed bound.

Case 3: $I = (\xi)$, where $\xi \notin \ell^1$. By (2), $I_{a_\infty} = \text{se}(\omega)$. We build a nonsummable $\eta \in c_0^*$ whose principal ideal omits ρ . Set $N_0 = 0$. Having chosen N_{k-1} , put $q_k = N_{k-1} + 1$ and choose $h_k > 0$ so that

$$h_k < \min \left\{ \frac{1}{k}, \frac{h_{k-1}}{2}, \frac{\rho_{kq_k}}{k} \right\} \quad (6)$$

for $k > 1$, omitting the middle term for $k = 1$. Let $L_k = \lceil 1/h_k \rceil$, set $N_k = N_{k-1} + L_k$, and put $\eta_j = h_k$ for $q_k \leq j \leq N_k$. Then η is nonincreasing and tends to zero, but

$$\sum_j \eta_j = \sum_k L_k h_k \geq \sum_k 1 = \infty.$$

Thus $J = (\eta)$ is not contained in the trace class, and (2) gives

$$J_{a_\infty} = \text{se}(\omega) = I_{a_\infty}.$$

It qualifies. If $\rho \leq CD_m\eta$, choose $k \geq m$ with $k > C$ and again evaluate at $n = mq_k$. Equation (6) yields

$$\rho_{mq_k} \geq \rho_{kq_k} > kh_k > C(D_m\eta)_{mq_k},$$

the same contradiction.

We have shown that every infinite-rank $\rho \in c_0^*$ is missing from at least one ideal in the defining intersection. Thus $\hat{I}^\infty \subset F$, and (3) proves equality. $\square$

5 Verification notes

The attempt was re-read adversarially against the exact source conventions. The current verdict is **likely valid** (confidence 93/100).

Check	Finding
All generator cases	Finite-rank, summable infinite-rank, and nonsummable cases are disjoint and exhaustive.
Tail domination	A suffix of a decreasing finite block has average at most the full-block average; future block masses are preserved exactly.
Ampliations	For fixed m, C , a stage $k \geq m, k > C$ contradicts the bound at mq_k . This explicitly handles the dilation freedom in principal-ideal membership.
Nonsummability	Each plateau has mass $L_k h_k \geq 1$, while $h_k \rightarrow 0$.
External input	Equation (2) is precisely [2, Lemma 4.7]; the principal-membership convention agrees with [1, §2.1].
Computation	None used or needed.

The most important specialist checks are the implication from (5) to $I_{a_\infty} \subset J_{a_\infty}$, the application of Lemma 4.7 in Case 3, and the direct finite-rank identity $F_{a_\infty} = F$.

6 Novelty search and limitations

On 9 August 2026, the lightweight registry, solution, attempt, and proof-gap indexes for the run were searched for arXiv:0707.3271, the exact Question 7 wording, and the core arithmetic-mean-at-infinity cancellation terms. The local parsed-source corpus was searched for notation and phrase variants. Bounded web searches used the exact question, variants of “ $\hat{I}^\infty$,” “arithmetic mean at infinity cancellation,” and the authors’ names. They returned the 2007 Kaftal–Weiss multipaper series and their later work on majorization, but no separate paper recording an answer to Question 7. This supports, but does not guarantee, novelty.

The theorem answers Question 7 exactly. It does not address the survey’s other nine questions or cancellation envelopes built from other arithmetic mean operations. The proof uses the known identity (2); it does not reprove that substantial background result.

References

- [1] V. Kaftal and G. Weiss, *A survey on the interplay between arithmetic mean ideals, traces, lattices of operator ideals, and an infinite Schur–Horn majorization theorem*, arXiv:0707.3271 (2007).
- [2] V. Kaftal and G. Weiss, *Traces on operator ideals and arithmetic means*, arXiv:0707.3169 (2007), especially Lemma 4.7.